FMI	**ADMINISTRATIVE INFORMATION**	**FMI No: 25350**

FMI Title: LightSpeed 5.x w/ Performix Pro (MCS_7079) Tube - Pressure Switch & Pump Replacement

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☐ Mandatory Safety	GEMS - AM:	Issue Date:
☑ Mandatory	GEMS - E:	
☐ Mandatory on Request	GEMS - A:	Close Date:

☑ **Staged FMI** — **Customer List** provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole.

☑ **Parts** and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.

☐ **Parts Ordering**: GEMS - Americas, Contact your Region logistics coordinator
GEMS - Europe, Order FMI kits from GPO/EPL Buc
GEMS - Asia, Contact your region logistics coordinator

☐ **No Parts** are required to complete this FMI.

☐ **Batch with** ____________________

To minimize costs, combine implementation of this FMI with the FMI's listed for all customers where the effectivity is the same (Please see your customer lists).

☐ **Non-Staged** — **No Customer List is provided**. Each service location must identify affected units, and place part orders as indicated in the Special Instructions.

Technical Support

GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems Insite Center (414) 524-5300 800-321-7973 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:

Unit Tracked:

FMI Instructions Part No.: 5126334-100

Do NOT leave unsecured on site

FMI Completion Certification Report

Mail To

GEMS - Americas	GEMS - Europe	
G.E. Medical Systems Attn: FMI Admin. W-523 P.O. Box 414 Milwaukee, WI 53201	GEMS - Europe FMI Admin. 322 BP34 F78533 Buc Cedex, France	Send to your Country FMI or Service Administrator

FMI No.: 25350

FMI Title: LightSpeed 5.x w/ Performix Pro (MCS_7079) Tube - Pressure Switch & Pump Replacement

Customer Full Name: ______________________

Number and Street Address: ______________________

City and Country: ______________________

____

Country Code   Site Number   System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not in area. Product Locator notified.

Customer Refusal or Equipment Altered - Modification Pending.

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Installer Comments ______________________

Effectivity

Costumer Name	Scanner Type	Sys ID
Molinette Italy	HP80	A5177358
HIA Val De Grace	HP80	M4434860
CED	HP80	A5269709
Centro Cardiologico Monzino (Milano)	HP80	A5624105
Pontificia Universidad Catolic (Santiago, Chile)	HP80	26329CT4
Ctr de Lutte Contre Cancer Rene Huguenin (St-Cloud, France)	HP80	M4143946
MIA. MATER HOSPITAL	RT	0910 21 2110
AUSTIN & REPAT. HOSP.	RT	0910 21 3079
MIA. JOHN FAUKNER HOSP.	RT	0910 21 3072

Purpose

The purpose of this FMI is to upgrade the Heat Exchanger and Coolant Pump. After these changes, an additional kit must be used to purge all air that may be present in the Heat Exchanger Coolant system.

Prerequisite FMIs

none

Related FMIs

none

Furnished Materials

Qty	Description	Part #
1	10 psi switch	5122148
1	Thumbscrew Clamp	5122103
1	Heat Exchanger Coolant Pump	5112969
1	Teflon Tape Roll	46-208982P1

Tool Required

Qty	Description	SW Version	Part #
1	FMI Document	N/a	5126334-100
1	Performix Pro (MCS_7079) Purge Kit	N/a	5122149
1	Permanent Marker - Black	N/a	N/a

People Power

"4" hours labor and "1.5" hours of travel

FMI Process Flow

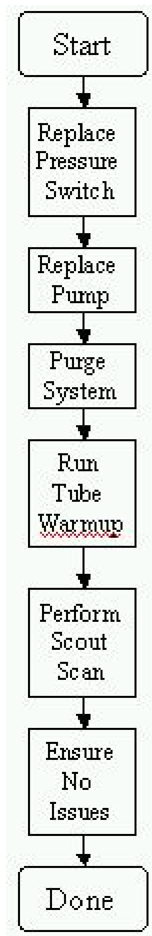


Figure 1: FMI Process Flow

Procedure

Preparation or Before you Begin

1. Read this entire document before you begin this procedure
2. Ensure the contents of the package are complete
3. Ensure that your system has a Performix Pro (MCS_7079) X-Ray tube. There are several ways to do this:
   a. Go to Reconfig and click on Hardware to look at tube type. The table below lists the possibilities

Software Version	LightSpeed 5x HP80 Tube Type	LightSpeed 5x RT Tube Type
M3	MX_240MCT	**MCS_7079**
04MW13.7	**OEM**	**OEM**

Table 1: Tube Configuration per system type as shown in Reconfig

Only the OEM and MCS_7079 selections correspond to the MCS_7079 tube

   b. Open the right side gantry cover and verify that the hoses are blue and are labeled "Water-Based Coolant"

Heat Exchanger Pressure Switch Replacement

1. Remove the Side, Top, and Front gantry covers.
2. Disable the Axial Drive power by turning off the AXIAL ENABLE switch on the Service Switch Board.

3. Position the heat exchanger so that the pressure switch is at the top of the heat exchanger. The surface that the pressure switch is threaded into should be as close to horizontal as possible.

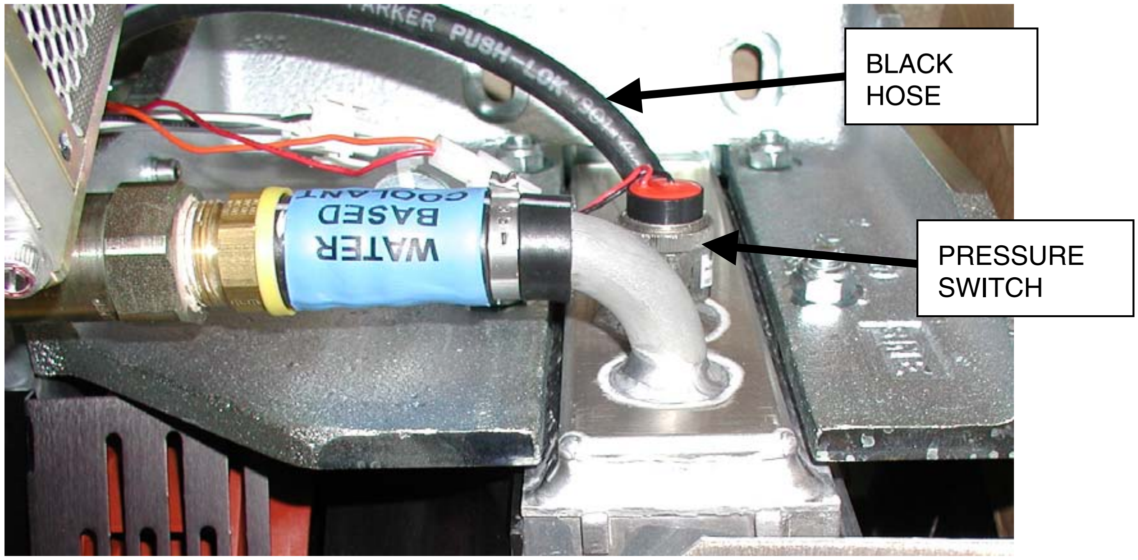


Figure 2: Oil Pressure Switch

4. Turn off all three service switches on the Service Switch Board (120VAC, Axial Drive Enable, HVDC Enable).
5. Engage gantry rotational lock. Verify that the gantry is locked by attempting to rotate the gantry by hand.
6. With the screw clamp provided, clamp the black hose so that no fluid can pass (see figure).

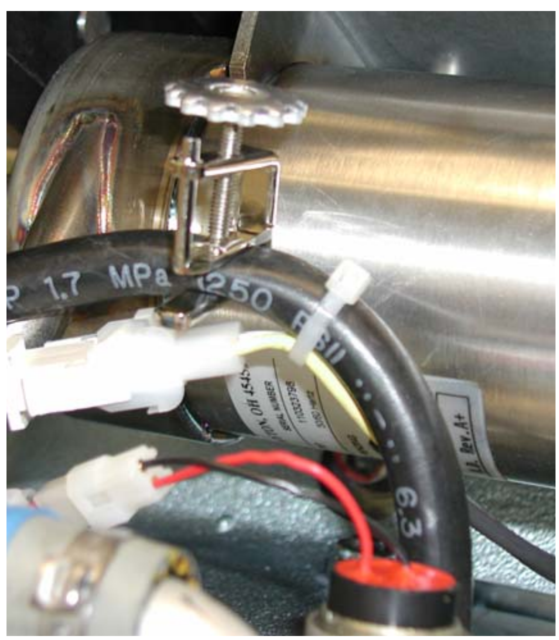


Figure 3: Black hose clamp in use

7. Prepare the new pressure switch with Teflon tape as shown the figure below.

Note: Do not cover the opening of the switch or the system will not work.

Figure 4: How to prepare the new oil pressure switch

8. Disconnect the wires connecting the heat exchanger to the pressure switch.
9. Wrap absorbent pad around the threads of the pressure switch on the heat exchanger.
10. Unscrew the pressure switch from the heat exchanger. Just before the switch is free go slowly to absorb any fluid.

Note: There should be no release of fluid from if the clamp is set properly on the black hose.

11. Install the new pressure switch finger tight and then turn an additional 1.5 turns with a wrench.
12. Connect the wires to the pressure switch at the quick disconnect.
13. Remove the clamp from the black hose.
14. Put any tie-wraps back in place to hold wires to the black hose. See figure 3.

NOTE: If you do not remove the clamp – the system will report overpressure errors via the JEDI generator.

15. Disengage gantry rotational lock.
16. With a marker, add a "-2" to the part number listed on the heat exchanger, such that the heat exchanger now indicates that it is part number 2383952-2.
17. Next you will need to replace the Pump, and then purge any air from the system.
18. Replace the Heat Exchanger Pump.
    a. See Appendix A or the System Service Manual for instructions if needed.
19. Purge Air from the Heat Exchanger Subsystem.
    a. See Appendix B for instructions if needed.
20. Run Tube Warmup

21. Perform scout scan
22. If no issues arise, safely restore the system.

Appendix A

Heat Exchanger Pump Replacement (Pro16)

1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Timing Not Applicable	30 mins	Timing Not Applicable

2 Overview

To replace the X-Ray heat exchanger pump:

- Remove old pump
- Mount new pump

Last Revised:	March 2004

3 Preliminary Requirements

3.1 Tools and Test Equipment

No tools or test equipment required.

3.2 Consumables

No consumables required.

3.3 Replacement Parts

No replacement parts required.

3.4 Safety

WARNING

POTENTIAL FOR SHOCK.

VOLTAGE MAY BE PRESENT.

USE PROPER LOCKOUT/TAGOUT PROCEDURES BEFORE REPLACING COMPONENTS.

3.5 Required Conditions

No required conditions.

4 Procedure

4.1 HX Pump Removal

WARNING

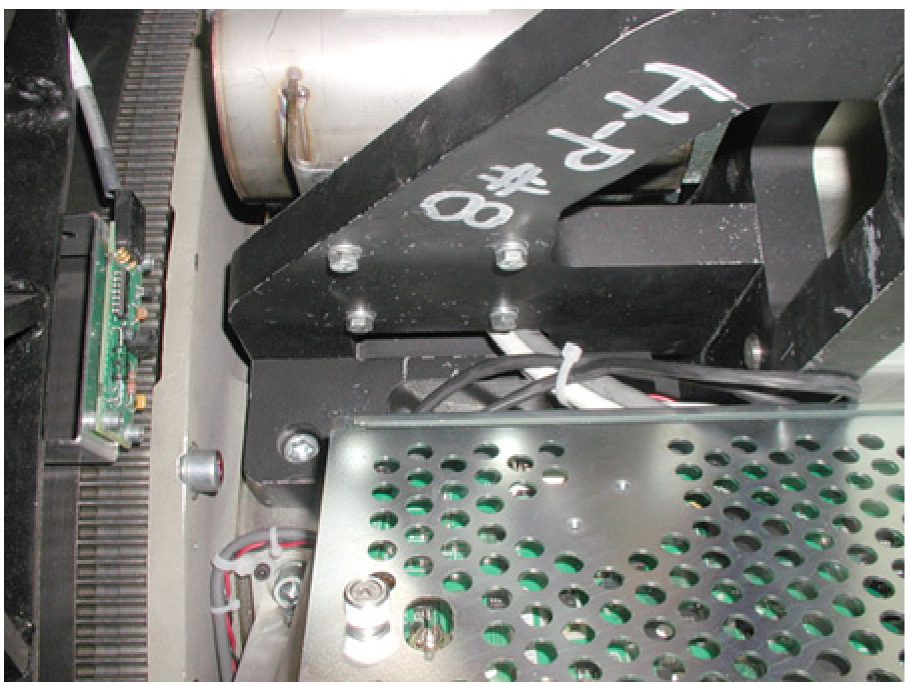
POTENTIAL FOR SHOCK.

VOLTAGE MAY BE PRESENT.

USE PROPER LOCKOUT/TAGOUT PROCEDURES BEFORE REPLACING COMPONENTS.

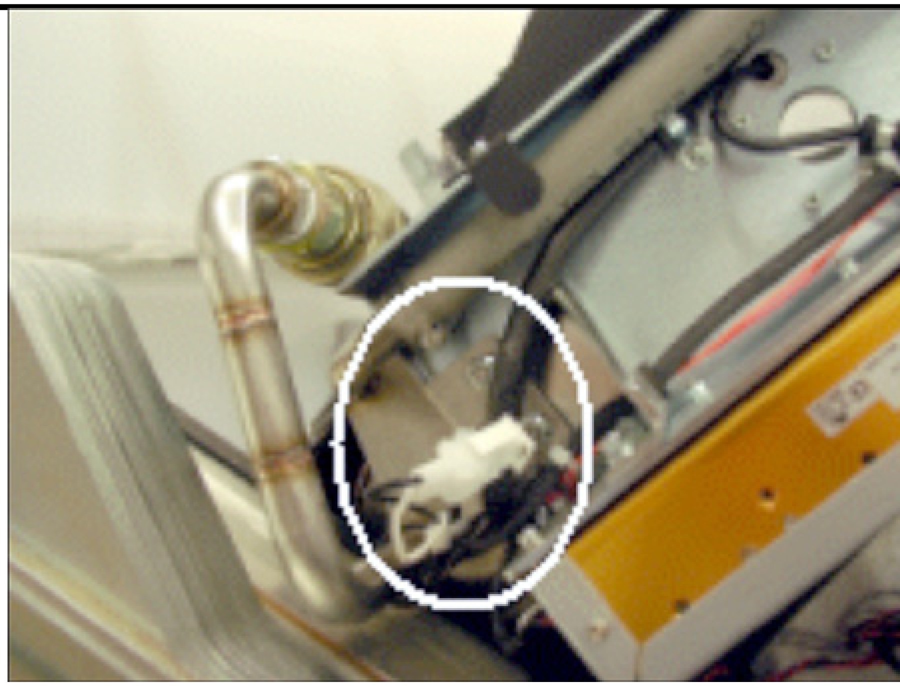
1. Remove gantry power. Be certain to follow appropriate Lockout/Tagout procedures.
2. Remove side and front gantry covers.
3. Turn OFF all three switches (**AXIAL DRIVE ENABLE**, **HVDC ENABLE**, **120 VAC**) on the Service Switch Panel.
4. Rotate the gantry so the tube is at the 2 o'clock position.
5. Remove the 4 M6 screws (Illustration 1) that mount the pump to the 107 weight stack bracket.

Illustration 1: HX pump mounting M6 screws



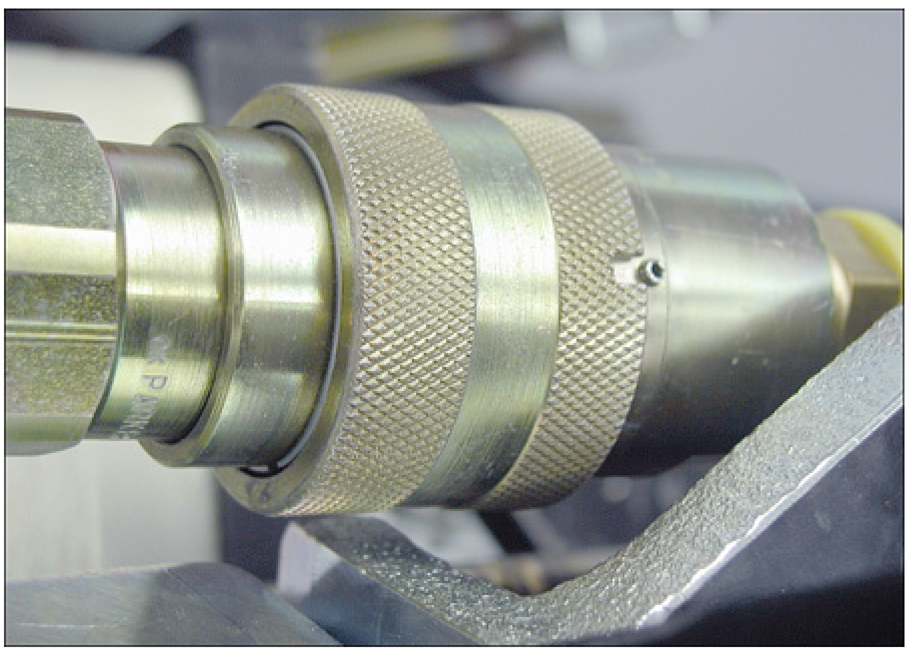
6. Disconnect HX Pump 120 VAC at quick disconnect (see Illustration 2).
7. Remove two M6 nuts (shown in Illustration 2 behind 120 VAC quick disconnect) for the pump's front mounting plate.

Illustration 2: 120VAC connector and front M6 nuts



8. Disconnect pump from heat exchanger at quick disconnect. Line the slot up with the pin and push. The pump should easily disengage (see Illustration 3).

Illustration 3: Align Pin and Slot

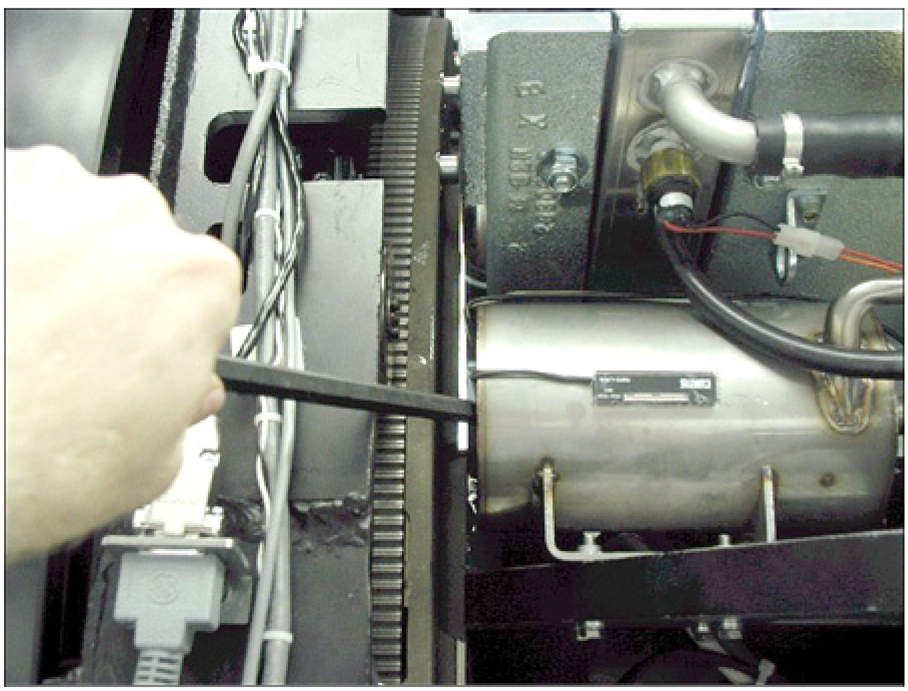


9. Disconnect oil hose from X-Ray tube.
10. Remove power interface board cover.
11. Remove pump and hose from gantry.

4.2 HX Pump Installation

1. Rotate gantry so that X-Ray tube is at the 2 o'clock position.
2. Connect pump to Heat Exchanger at quick disconnect. Make sure you align the pin and slot, then connect. See Illustration 3 in HX Pump Removal.
3. Route the hose to the tube. It routes underneath the Heat Exchanger shroud, but between the shroud and the clear plastic power interface board cover.
4. Attach quick disconnect and safety mechanism at tube.
5. Reattach the Power Interface board cover.
6. Connect pump 120 VAC connector.
7. Unmount the ORP assembly by removing the two M12 bolts.
8. Install the 4 M6 HX pump mounting screws. If you just replaced the heat exchanger, it may require some force to realign the pump. Using a lever helps but be sure not to damage the hour meter. See Illustration 4.

Illustration 4: Using lever to push pump around to align holes



9. Attach the ORP assembly.
   a. Apply "pre-load" torque to the two M12 bolts:

25 ft.-lbs	33 N-m	295 in-lbs	340 kg-cm

   b. Apply final torque to the two M12 bolts:

49 ft.-lbs	66 N-m	590 in-lbs	680 kg-cm

10. Attach the two M6 nuts for the front pump.

5 Finalization

1. Ensure that proper torque specifications (see *Torque Wrench Information*) are followed for all fasteners.

Appendix B – Heat Exchanger Air Purging

Objective of procedure

The objective of this procedure is to purge air from the liquid tube cooling system. High amounts of air present in the system can lead to a decrease in flow which can then result in potential tube overheating / over pressure problems and / or image quality problems.

Estimated Time to Complete

45 minutes

Procedure summary

Power down gantry

Remove air from purge kit hoses

Attach purge kit between heat exchanger and x-ray tube

Turn on gantry 120VAC to enable heat exchanger pump

Keep power on until there is no milky colored liquid in purge kit

Remove purge kit

Safely restore system

Special Tools

Performix Pro Purge Kit (Part number 5122149)

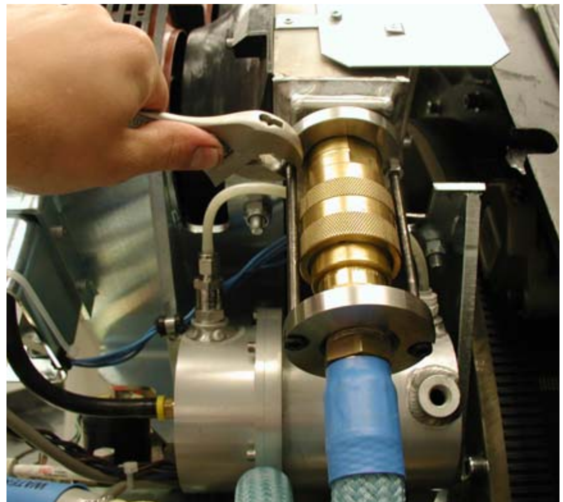
Procedure Details

GANTRY PREP:

1. Remove side, and top gantry covers.
2. Turn off all 3 service switched on the Service Switch Board (120 VAC, Axial Drive Enable, HVDC Enable)
3. Remove front Gantry Cover. Cable disconnection is optional based on amount of space desired for procedure.

4. Position Gantry such that quick disconnect on blue hose (between tube and Heat Exchanger) is easily accessible.

5. Loosen Jam Nuts at bottom of quick disconnect bolts and remove the bolts. Do NOT disconnect the hose at this point.



6.

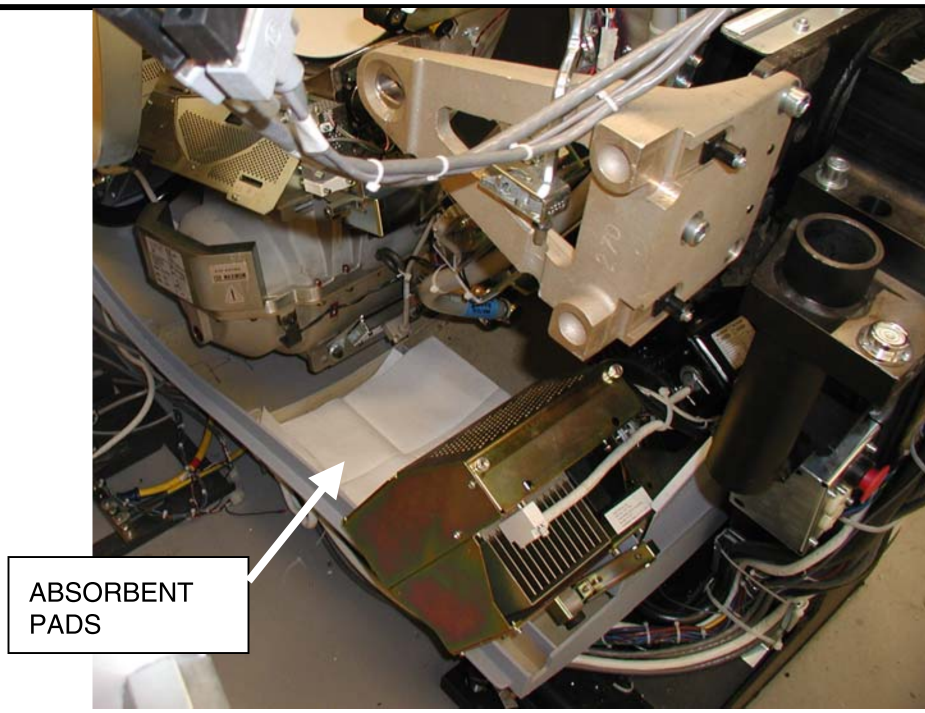
7. Rotate the gantry such that the tube is in between the 5 and 6 o'clock positions and place absorbent pads in base of cover under the hose (shown in figure).

CAUTION: Pinch Point

The following work to be performed could create inadvertent gantry rotation causing injury

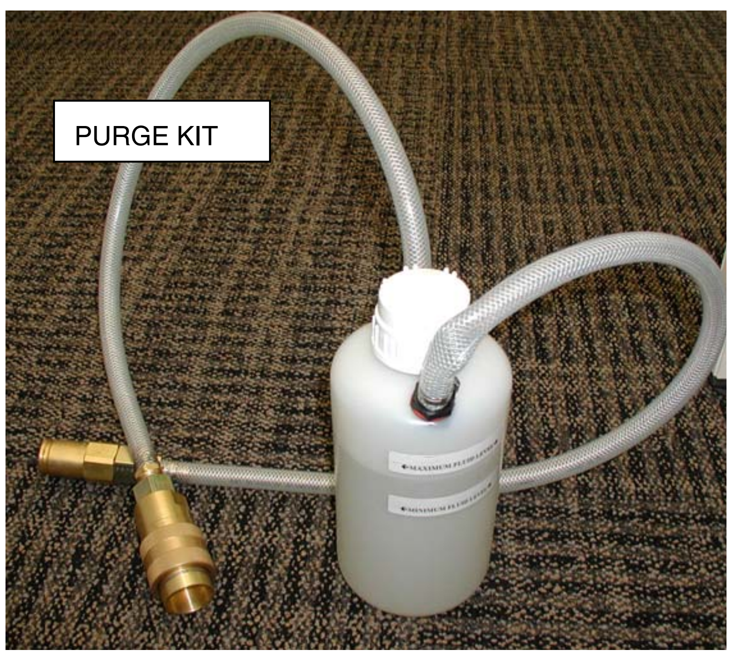
Engage gantry rotational lock to prevent gantry rotation

8. Engage gantry rotational lock. Ensure that the gantry is locked by attempting to rotate the gantry by hand.



PURGE BOTTLE PREP:

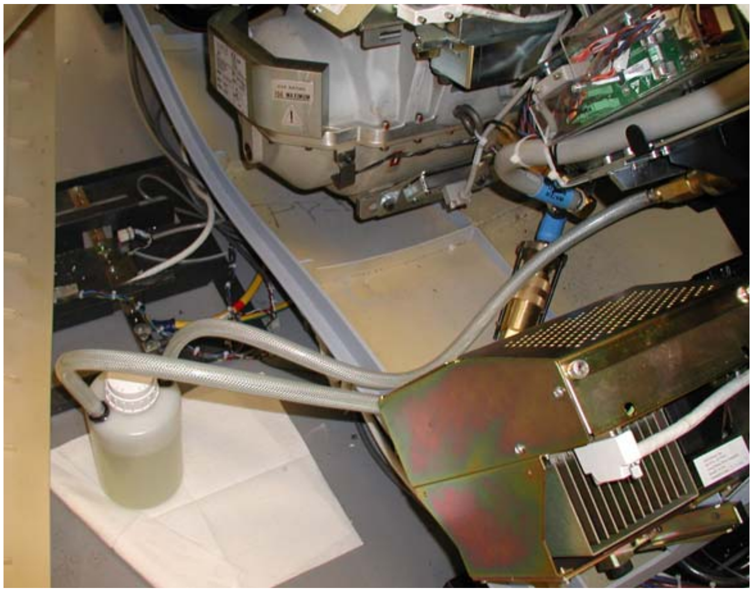
1. Unpack the purge kit



2. Connect the quick disconnects on the purge kit together.
3. Make sure the cap is tight.
4. Turn the bottle upside down until all air in the hoses is removed. Verify that all air is removed from hoses by noting lack of bubbles flowing into the bottle.
5. With the bottle still inverted, disconnect the hoses from each other.
6. Return the bottle to the upright position.

ATTACH PURGE KIT TO GANTRY:

1. Disconnect the hose between the heat exchanger and the tube.
2. Connect the purge kit between the tube and the heat exchanger and set it on absorbent pads on the floor.



3. Loosen the cap on the bottle 1 complete turn to allow for venting during the purge process.

PURGE PROCEDURE:

1. Turn on the Heat Exchanger pump by turning on the Gantry 120 VAC switch on the service switch board.
2. Watch for the following:
   a. LEAKS – if any leaks are detected, turn off the 120VAC switch and repair the leak before continuing
   b. FLUID COLOR CHANGE – If there is any air in the system, the fluid in the purge kit will become a milky color. If there is no air in the system, the fluid will remain a clear lime green color

c. FLUID LEVEL – During purging the fluid level in the bottle may rise and fall approximately 5cm. If the fluid level rises to the top of the bottle, remove enough fluid from the bottle to ensure that the bottle does not overfill.

3. Once the fluid is a clear color (not milky colored) AND the fluid level in the bottle has stabilized turn off the gantry 120VAC to remove power from the heat exchanger pump.

Note: This will usually happen either immediately (indicating no air was present in system before purging), or after some period of time (fluid turns milky color then becomes clear again after approximately 10-15 minutes)

4. Tighten the cap on the purge bottle
5. Elevate the purge bottle so that it is higher than the hose connections.
6. At this height above the hose connections, hold the purge bottle upside-down so that the fluid can drain into the heat exchanger's accumulator.
7. Hold purge bottle in this inverted position for approximately 15-45 seconds.
8. Return purge bottle to upright position on the floor.
9. Disconnect the purge kit from the heat exchanger and tube.
10. Reconnect the tube to the heat exchanger
11. Disengage gantry rotational lock.
12. Rotate the gantry such that the quick disconnects for the hose are easily accessible.
13. Tighten bolts on quick disconnect.
14. Snug the jam nuts on the quick disconnect.
15. Safely restore system for customer use.

Finalization

Run Tube warmup and verify no errors.